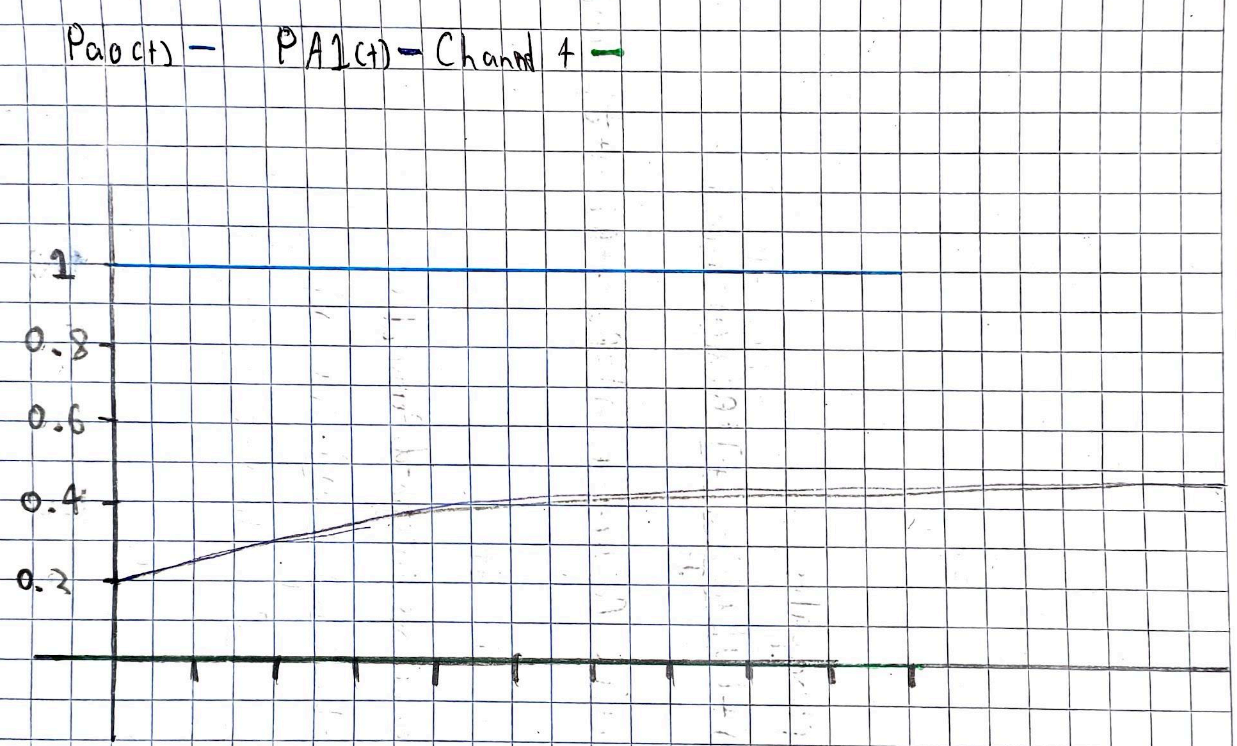
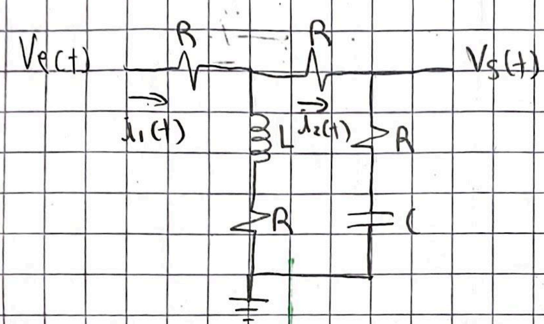

Pa0(t) –
PA1(t) – Channel 4 –
1
0.8
0.6
0.4
0.2



Ecuaciones principales

$$2Ri_1(t) \qquad Ri_1(t) - Ri_2(t)$$

$$V_e(t) = Ri_1(t) + L\frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L\frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = Ri_2(t) + Ri_2(t) + \frac{1}{C}\int i_2(t)\,dt$$

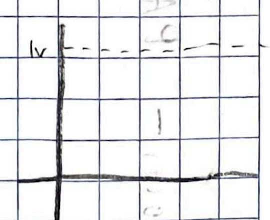
$$V_s(t) = Ri_2(t) + \frac{1}{C}\int i_2(t)\,dt$$

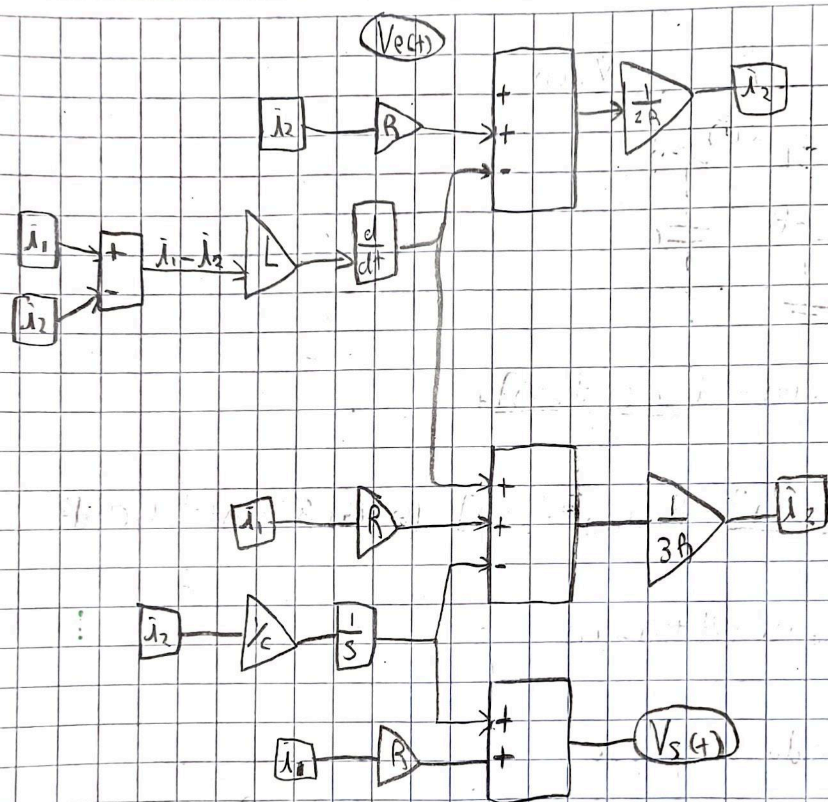
Modelo de ecuaciones integro-diferenciales

$$i_1(t) = \left[V_e(t) - L\frac{d[i_1(t) - i_2(t)]}{dt} + Ri_2(t)\right]\frac{1}{2R}$$

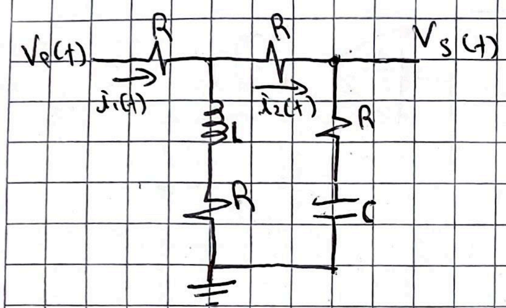
$$i_2(t) = \left[L\frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C}\int i_2(t)\,dt\right]\frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C}\int i_2(t)\,dt$$





Ve(t)
$\bar{i}_2$
R
$\frac{1}{2R}$
$\bar{i}_2$
$\bar{i}_1$
$\bar{i}_2$
$\bar{i}_1 - \bar{i}_2$
L
$\frac{d}{dt}$
$\bar{i}_1$
R
$\frac{1}{3R}$
$\bar{i}_2$
$\bar{i}_2$
$1/c$
$\frac{1}{s}$
$\bar{i}_1$
R
$V_s(t)$



$$V_e(t) = R\,i_1(t) + L\frac{d[i_1(t) - i_2(t)]}{dt} +$$

$$L\frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R\,i_2(t)\;R\,i_2(t) + \frac{1}{C}\int i_2(t)\,dt$$

$$V_s(t) = \frac{1}{C}\int i_2(t)\,dt + R\,i_2(t)$$

Transformada de Laplace

$$V_e(s) = R\,I_1(s) + LS[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)]$$

$$LS[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = R\,I_2(s) + R\,I_2(s) + \frac{I_2(s)}{CS}$$

$$V_s(s) = R\,I_2(s) + \frac{I_2(s}{CS}$$

Procedimiento algebraico

$$V_{e(s)} = (R + LS + R)I_{1(s)} - (LS + R)I_{2(s)}$$
$$= (LS + 2R)I_{1(s)} - (LS + R)I_{2(s)}$$

$$LSI_{1(s)} - LSI_{2(s)} + RI_{1(s)} - RI_{2(s)} = 2RI_{2(s)} + \frac{I_{2(s)}}{CS}$$

$$LSI_{1(s)} + RI_{1(s)} = 3RI_{2(s)} + LSI_2 + \frac{I_{2(s)}}{CS}$$

$$(LS + R)I_{1(s)} = (3R + LS + \frac{1}{CS})I_{2(s)}$$

$$I_{1(s)} = \frac{3CRS + CLS^2 + 1}{CS(LS + R)} I_{2(s)} = \frac{CLS^2 + 3CRS + 1}{CS(LS + R)} I_{2(s)}$$

$$V_{e(s)} = \frac{(LS + 2R)(CLS^2 + 3CRS + 1)}{CS(LS + R)} I_{2(s)} - (LS + R)I_{2(s)}$$

$$= \left[\frac{(LS + 2R)(CLS^2 + 3CRS + 1) - CS(LS + R)(LS + R)}{CS(LS + R)}\right] I_{2(s)}$$

$$CL^2S^3 + 3CLRS^2 + LS + 2CLRS^2 + 6CR^2S + 2R$$

$$- CL^2S^3 - 2CLRS^2 - CR^2S$$

R = 3.3E3
L = 1.5E-3
C = 4.7E-6

$$V_{e(s)} = \frac{3CLRS^2 + (5CR^2 + L)S + 2R}{CS(LS + R)}$$

$$V_{s(s)} = \frac{\frac{CRS + 1}{CS} I_{2(s)}}{\frac{3CLRS^2 + (5CR^2 + L)S + 2R}{CS(LS + R)}}$$

$$\frac{V_{s(s)}}{V_{e(s)}} = \frac{CLRS^2 + (CR^2 + L)S + R}{3CLRS^2 + (5CR^2 + L)S + 2R}$$

$$(CRS + 1)(LS + R) = CLRS^2 + CR^2S + LS + R$$

num = [(4.7E-6)*(3.3E-3)*(3.3E3),(4.7E-6)*(3.3E3)^2+3.3E-3, 3.3E3

Estabilidad en lazo abierto

Calcular los polos de la función de transferencia

$$\frac{V_{s(s)}}{V_{e(s)}} = \frac{CLRS^2 + C(CR^2+L)S + R}{3CLRS^2 + (5C(R^2+L)\cdot S + 2R}$$

den = [3*C*L*R, 5*C*R**2+L, 2*R]

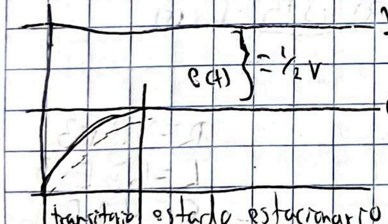
L = np.roots(den

→ fprint: Las raices son {L[0]} y {L[1]}

$\lambda_1 = -3{,}666{,}662.368$

$\lambda_2 = -25.790$

El sistema presenta una respuesta estable y sobreamortiguada



$V_{e(t)} = 1\,V$

$V_{e(s)} = 1/s$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s \cdot V_{e(s)} \left[1 - \frac{V_{s(s)}}{V_{e(s)}}\right]$$

$$= \lim_{s \to 0} s \cdot \frac{1}{s} \left[1 - \frac{CLRS^2 + C(R^2+L)S + R}{3CLRS^2 + (5C(R^2+L))S + 2R}\right]$$

$$= \frac{R}{2R}$$

$e(t) = 1/2\ V$